

Lightning Detector - Raspberry Pi Flash and Deploy Guide

This guide is generic and site-agnostic.

1. Flash Raspberry Pi OS Lite (64-bit) using Raspberry Pi Imager.
2. In advanced settings, set hostname, SSH, username/password, WiFi, timezone.
3. Boot the Pi and connect via SSH.
4. Enable SPI: `sudo raspi-config nonint do_spi 0`
5. Install packages: `sudo apt update; sudo apt upgrade -y; sudo apt install -y python3-pip python3-venv pigpio`
6. Enable pigpio daemon: `sudo systemctl enable pigpiod; sudo systemctl start pigpiod`
7. Copy project files to `/home/pi`.
8. Install Python dependencies: `pip3 install --break-system-packages -r requirements.txt`
9. Create data folder: `mkdir -p /home/pi/lightning_data`
10. Run calibration script and set `tune_cap` in `lightning_config.json`.
11. Install service file, reload systemd, enable and start service.
12. Verify operation with `systemctl status` and `journalctl` logs.

Recommended generic defaults:

- `station_id`: GENERIC-LD-001
- `stratus_endpoint`: https://stratusweather.co.za/api/ingest/YOUR_INGEST_ID

Campbell wiring:

- GPIO 26 -> C2 (serial RX)
- GPIO 19 -> P1 (pulse input)
- GND -> G